

CU BOULDER / COURSERA

Spacecraft Dynamics Capstone Project

Attitude Dynamics and Control of a Nano-Satellite
Orbiting Mars

Course Report

Student: Haniyeh Zahra Budaqi

Instructor: Prof. Dr. Hanspeter Schaub

May 2026

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Abstract

In this project, the knowledge acquired from the Spacecraft Dynamics and Control specialization is applied to perform attitude control of a nano-satellite orbiting Mars while operating as a daughter-craft to a mother spacecraft in a geostationary Mars orbit (GMO). Through the completion and analysis of eleven project tasks, an attitude control architecture is developed and validated. The resulting simulation enables three operational modes based on the spacecraft mission requirements and relative orbital geometry:

- 1) Science mode: pointing the spacecraft sensor toward the Martian surface,
- 2) Power mode: orienting the solar panels toward the Sun,
- 3) Communication mode: pointing the communication antenna toward the mother spacecraft.

The spacecraft dynamics, reference-frame generation, attitude error evaluation, and closed-loop control simulations are implemented in MATLAB, while mission visualization is performed using AGI Systems Tool Kit (STK).

Chapter 1

Introduction

This project considers a small science satellite orbiting Mars in a Low Mars Orbit (LMO), tasked with collecting scientific data from the Martian surface and relaying the acquired information to a larger mother spacecraft orbiting in a Geostationary Mars Orbit (GMO). In addition to science and communication operations, the spacecraft must periodically orient its solar panels toward the Sun to maintain sufficient onboard power.

Accordingly, the spacecraft mission is divided into three operational modes:

- 1) **Science Mode:** pointing the sensor toward the Martian surface,
- 2) **Power Mode:** orienting the solar panels toward the Sun,
- 3) **Communication Mode:** pointing the communication antenna toward the GMO spacecraft.

Both the LMO and GMO spacecraft are assumed to follow known circular orbits whose orientations are described using Hill reference frames generated through a 3-1-3 Euler-angle sequence.

The LMO spacecraft is equipped with:

- A science sensor aligned with the $+b_1$ body axis,
- Solar panels whose normal vector aligns with the b_3 axis,
- A communication antenna aligned with the $-b_1$ axis.

The objective of this project is to design and validate an attitude determination and control framework capable of autonomously switching between the required operational pointing modes.

Chapter 2

Mission Scenario Definitions

2.1 Mission Orbit Overview

The LMO and GMO spacecraft are both assumed to follow circular orbits around Mars. The orbital motion is modeled using classical two-body dynamics, where Mars is treated as the central gravitational body.

The physical and orbital parameters used throughout the simulations are summarized in Table 2.1.

Table 2.1: Simulation and Spacecraft Parameters

Parameter	Value	Unit
Mars radius R_{Mars}	3396.19	km
Mars gravitational parameter μ_{Mars}	4.28283×10^{13}	m^3/s^2
LMO altitude	400	km
LMO orbital radius r_{LMO}	3796.19	km
GMO orbital radius r_{GMO}	20424.2	km
LMO orbital angular rate $\dot{\theta}_{LMO}$	0.000884797	rad/s
GMO orbital angular rate $\dot{\theta}_{GMO}$	0.0000709003	rad/s

The orbital angular rates are computed using the circular-orbit relation:

$$\dot{\theta} = \sqrt{\frac{\mu_{Mars}}{r^3}} \quad (2.1)$$

The initial orbit orientation angles of both spacecraft are summarized in Table 2.2.

Table 2.2: Initial Orbit Frame Orientation Angles

Spacecraft	Ω (deg)	i (deg)	$\theta(t_0)$ (deg)
LMO	20	30	60
GMO	0	0	250

Figure 2.1 illustrates the definition of a circular orbit using relative elements along with inertial and Hill frames' base vectors. Note that the orbital elements (Ω, i, θ) can be used as (3-1-3) Euler Angles to obtain Hill frame $\mathcal{H}$ from inertial frame $\mathcal{N}$.

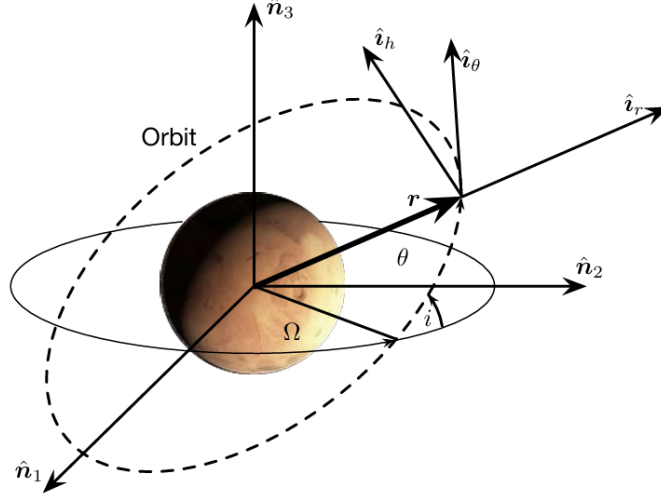


Figure 2.1: Illustration of circular orbital elements alongside inertial frame $\mathcal{N} : \{\hat{n}_1, \hat{n}_2, \hat{n}_3\}$ and Hill frame $\mathcal{H} : \{\hat{i}_r, \hat{i}_\theta, \hat{i}_h\}$ base vectors

2.2 Mission Pointing Scenarios

A body frame as shown in Figure 2.2 is mounted on the spacecraft, assumed to be the principal body frame.

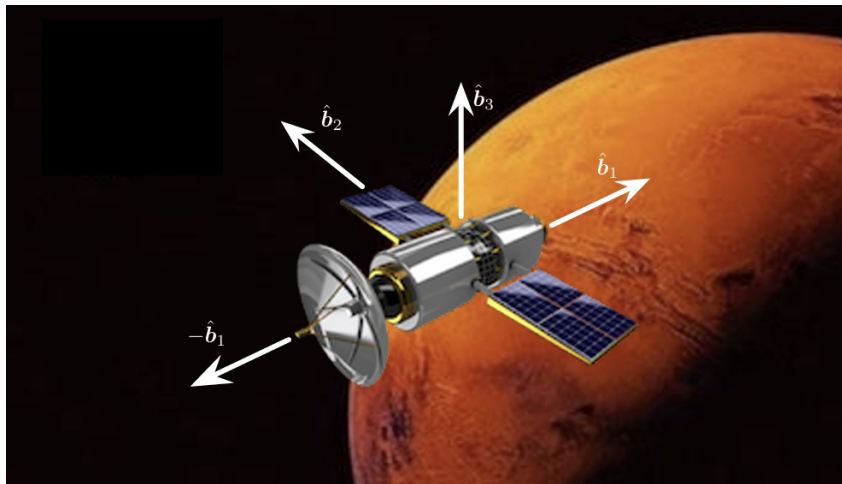


Figure 2.2: Illustration of LMO spacecraft's body frame $\mathcal{B} : \{\hat{b}_1, \hat{b}_2, \hat{b}_3\}$

Three operational spacecraft pointing modes are defined according to mission requirements.

The Sun is assumed to be located at an effectively infinite distance and aligned with the inertial n_2 axis. Therefore, during Sun-pointing operation, the spacecraft solar-panel axis b_3 must align with the inertial n_2 direction.

The spacecraft operational modes are summarized as follows:

- **Power Mode:** active whenever the spacecraft is on the sunlit side of Mars,
- **Science Mode:** active during eclipse when GMO communication is unavailable,
- **Communication Mode:** active whenever the GMO spacecraft lies within the communication line-of-sight constraint.

The communication mode is activated when the angular separation between the LMO and GMO position vectors is less than 35° .

This operational logic prioritizes spacecraft power generation whenever solar illumination is available, while eclipse periods are allocated to science or communication operations.

Table 2.3: Mission Pointing Scenario Summary

Orbital Situation	Operational Mode
Spacecraft on sunlit Mars side	Power Mode
Spacecraft in eclipse and GMO visible	Communication Mode
Spacecraft in eclipse and GMO not visible	Science Mode

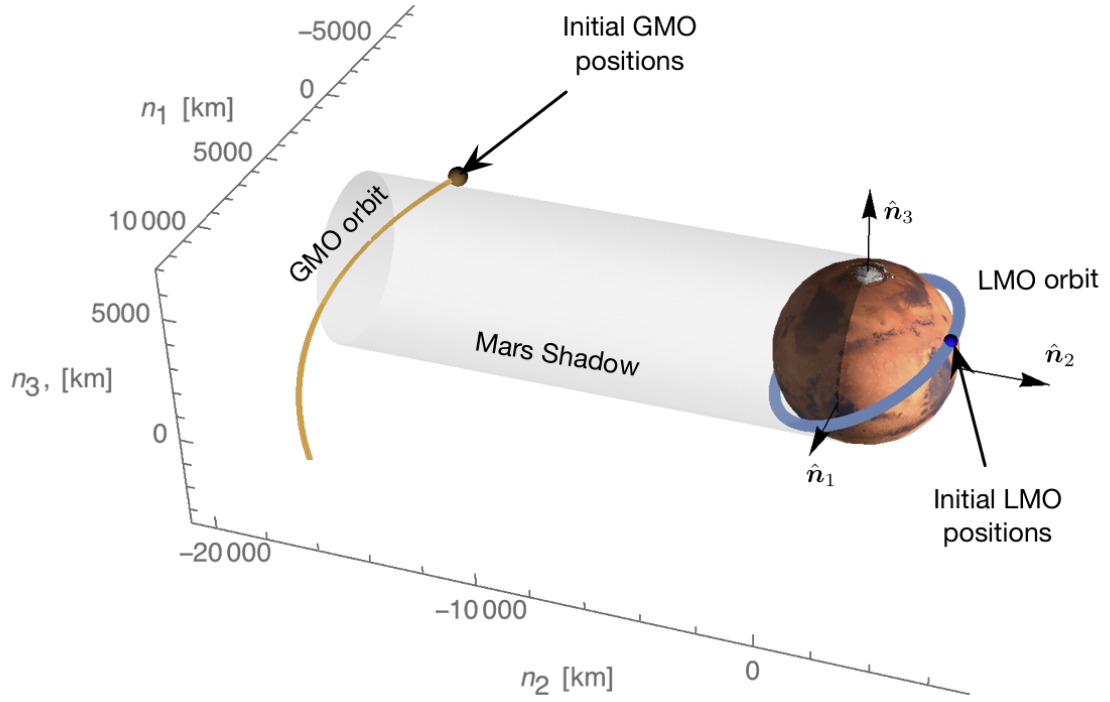


Figure 2.3: Mission Orbit Scenario Illustration

2.3 Spacecraft Attitude States

The spacecraft initial attitude is represented using Modified Rodrigues Parameters (MRPs), while the initial angular velocity is expressed in the spacecraft body frame.

The initial spacecraft attitude is defined as:

$$\sigma_{B/N}(t_0) = \begin{bmatrix} 0.3 \\ -0.4 \\ 0.5 \end{bmatrix}$$

The initial body angular velocity is:

$${}^B\omega_{B/N}(t_0) = \begin{bmatrix} 1.00 \\ 1.75 \\ -2.20 \end{bmatrix} \text{ deg/s}$$

The spacecraft inertia tensor is assumed diagonal, implying alignment between the body frame and principal inertia axes.

$${}^B[I] = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 7.5 \end{bmatrix} \text{ kg}\cdot\text{m}^2$$

The spacecraft is assumed to be capable of generating arbitrary three-axis control torques through an onboard actuator system.

2.4 Attitude Control Law Overview

A proportional-derivative (PD) control law is used throughout this project to regulate spacecraft attitude motion:

$${}^B\mathbf{u} = -K\boldsymbol{\sigma}_{B/R} - P{}^B\boldsymbol{\omega}_{B/R} \quad (2.2)$$

where:

- $\boldsymbol{\sigma}_{B/R}$ is the attitude tracking error,
- ${}^B\boldsymbol{\omega}_{B/R}$ is the angular velocity tracking error,
- K and P are controller gain matrices.

The controller objective is to asymptotically reduce attitude and angular velocity tracking errors while ensuring stable closed-loop rotational behavior.

2.5 Implementation Methods

[NEED REVISION BEFORE EVEN BEING WRITTEN]

Chapter 3

Task Descriptions and Solutions

3.1 Orbits

3.1.1 Task 1: Orbit Simulation

Assume the general orbit frame as $\mathcal{O} : \{\hat{i}_r, \hat{i}_\theta, \hat{i}_h\}$. Even though the general orbit frame and Hill frame basically use the same base vectors, their difference is found in their origin. $\mathcal{O}$ frame has its origin aligned with the origin of inertial frame $\mathcal{N}$ – which is the center of Mars. The Hill frame $\mathcal{H}$, however, is mounted on the spacecraft, therefore has the same origin as the body frame $\mathcal{B}$.

Both these frames are obtainable through a (3-1-3) Euler Angle rotation on the inertial frame, though the hill frame origin is also translated to the spacecraft.

Now, using the $\mathcal{O}$ frame, spacecraft position vector is expressed as:

$${}^{\mathcal{O}}\mathbf{r} = r\hat{i}_r$$

Both LMO and GMO orbits are circular, therefore having the same angular velocity $\dot{\theta}$ all over the orbit. Spacecraft velocity vector is expressed as:

$${}^{\mathcal{O}}\dot{\mathbf{r}} = r\dot{\theta}\hat{i}_\theta$$

The MATLAB function `orbitTheta()` takes time, angular velocity and initial angle to calculate the element θ using the following equation:

$$\theta_t = \dot{\theta}t + \theta_0$$

Taking the position and velocity vectors from the general orbit frame to the inertial frame is done using the DCM $[NO]$ which can be obtained having all orbital elements at a given time.

$$[ON] = M_3(\theta_t)M_1(i)M_3(\Omega)$$

$$[NO] = [ON]^T$$

$${}^{\mathcal{N}}\mathbf{r} = [NO] {}^{\mathcal{O}}\mathbf{r}$$

$${}^{\mathcal{N}}\dot{\mathbf{r}} = [NO] {}^{\mathcal{O}}\dot{\mathbf{r}}$$

This is achieved by the MATLAB function `orbit_state()`.

3.1.2 Task 2: Orbit Frame Orientation

The Hill-frame basis vectors were constructed directly from the inertial position and velocity vectors:

$${}^{\mathcal{N}}\hat{\mathbf{i}}_r = \frac{\mathbf{r}_{LMO}}{\|\mathbf{r}_{LMO}\|}$$

$${}^{\mathcal{N}}\hat{\mathbf{i}}_h = \frac{\mathbf{r}_{LMO} \times \dot{\mathbf{r}}_{LMO}}{\|\mathbf{r}_{LMO} \times \dot{\mathbf{r}}_{LMO}\|}$$

$${}^{\mathcal{N}}\hat{\mathbf{i}}_\theta = \hat{\mathbf{i}}_h \times \hat{\mathbf{i}}_r$$

To construct a DCM with the base vectors in either frames, we have:

$$[FB] = \begin{bmatrix} {}^{\mathcal{F}}\hat{\mathbf{b}}_1 & {}^{\mathcal{F}}\hat{\mathbf{b}}_2 & {}^{\mathcal{F}}\hat{\mathbf{b}}_3 \end{bmatrix} = \begin{bmatrix} {}^{\mathcal{B}}\hat{\mathbf{f}}_1^T \\ {}^{\mathcal{B}}\hat{\mathbf{f}}_2^T \\ {}^{\mathcal{B}}\hat{\mathbf{f}}_3^T \end{bmatrix}$$

Therefore, to find $[HN]$ we will have:

$$[HN] = \begin{bmatrix} {}^{\mathcal{N}}\hat{\mathbf{i}}_r^T \\ {}^{\mathcal{N}}\hat{\mathbf{i}}_\theta^T \\ {}^{\mathcal{N}}\hat{\mathbf{i}}_h^T \end{bmatrix}$$

A MATLAB function named `orbit_frame_dcm(t)` was implemented to evaluate the time-varying Hill-frame orientation.

3.2 Reference Frame Orientation

3.2.1 Task 3: Sun-Pointing Reference Frame Orientation

To point the spacecraft solar panels axis $\hat{\mathbf{b}}_3$ at the Sun, the reference frame $\mathcal{R}_s$ must be chosen such that $\hat{\mathbf{r}}_3$ axis points in the Sun direction ($\hat{\mathbf{n}}_2$ in the scenario). To define the

frame completely, the first axis $\hat{\mathbf{r}}_1$ is assumed to point in $-\hat{\mathbf{n}}_1$ direction. Therefore, the base vectors will be:

$$\begin{aligned}\hat{\mathbf{r}}_1 &= -\hat{\mathbf{n}}_1 \\ \hat{\mathbf{r}}_3 &= \hat{\mathbf{n}}_2 \\ \hat{\mathbf{r}}_2 &= \hat{\mathbf{r}}_3 \times \hat{\mathbf{r}}_1 = -(\hat{\mathbf{n}}_2 \times \hat{\mathbf{n}}_1) = \hat{\mathbf{n}}_3\end{aligned}$$

Similar to the previous task, the analytical expression for the DCM can be obtained using the base vectors:

$$[R_s N] = \begin{bmatrix} \mathcal{N} \hat{\mathbf{r}}_1^T \\ \mathcal{N} \hat{\mathbf{r}}_2^T \\ \mathcal{N} \hat{\mathbf{r}}_3^T \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

The resulting reference-frame DCM can also be obtained using a 3-1-3 Euler-angle of $(\pi, \frac{\pi}{2}, 0)$ rotation on it. MATLAB function `RsN_DCM()` serves this purpose.

Since the base vectors of this frame are just a fixed permutation of inertial frame basis vectors, the frame remains fixed relative to the inertial frame, thus:

$$\boldsymbol{\omega}_{R_s/N} = 0$$

3.2.2 Task 4: Nadir-Pointing Reference Frame Orientation

To point the spacecraft sensor platform axis $\hat{\mathbf{b}}_1$ towards the center of Mars (nadir direction), the reference frame $\mathcal{R}_n$ must be chosen such that $\hat{\mathbf{r}}_1$ axis points towards the planet. To completely describe the frame, we assume the second axis $\hat{\mathbf{r}}_2$ points in velocity direction $\hat{\mathbf{i}}_\theta$. Therefore, we'll have:

$$\begin{aligned}\hat{\mathbf{r}}_1 &= -\hat{\mathbf{i}}_r \\ \hat{\mathbf{r}}_2 &= \hat{\mathbf{i}}_\theta \\ \hat{\mathbf{r}}_3 &= \hat{\mathbf{r}}_1 \times \hat{\mathbf{r}}_2 = -\hat{\mathbf{i}}_h\end{aligned}$$

This frame can be obtained using a (3-1-3) Euler Angles rotation on the Hill frame with angles $(\pi, \pi, 0)$.

$$[R_n H] = M_1(\pi)M_3(\pi)$$

We also had previously obtained $[HN]$ in task 2 as a function of time. So,

$$[R_n N](t) = [R_n H][HN](t)$$

In MATLAB, this task is done by function `RnN_DCM()`.

Since this frame is a re-arrangement of the Hill frame, it also moves with it, thus:

$$\boldsymbol{\omega}_{R_n/N} = \boldsymbol{\omega}_{H/N} = \dot{\theta} \hat{\mathbf{i}}_\theta$$

3.2.3 Task 5: GMO-Pointing Reference Frame Orientation

To point the nano-satellite communication platform axis $\hat{\mathbf{b}}_1$ towards the GMO mother spacecraft, the communication reference frame $\mathcal{R}_c$ must be chosen such that $-\hat{\mathbf{r}}_1$ axis points towards the GMO satellite location, defining $\Delta \mathbf{r} = \mathbf{r}_{GMO} - \mathbf{r}_{LMO}$. So,

$$\hat{\mathbf{r}}_1 = -\frac{\Delta \mathbf{r}}{\|\Delta \mathbf{r}\|}$$

Further, to fully define a three-dimensional reference frame, assume the second axis is defined as

$$\hat{\mathbf{r}}_2 = \frac{\Delta \mathbf{r} \times \hat{\mathbf{n}}_3}{\|\Delta \mathbf{r} \times \hat{\mathbf{n}}_3\|}$$

The third basis vector is then obtained from

$$\hat{\mathbf{r}}_3 = \hat{\mathbf{r}}_1 \times \hat{\mathbf{r}}_2$$

Having all three base vectors in the inertial frame, we have:

$$[R_c N] = \begin{bmatrix} \mathcal{N} \hat{\mathbf{r}}_1^T \\ \mathcal{N} \hat{\mathbf{r}}_2^T \\ \mathcal{N} \hat{\mathbf{r}}_3^T \end{bmatrix}$$

The MATLAB function `RcN_DCM(t)` is written to deliver this task.

To derive the relative angular velocity ${}^{\mathcal{N}}\boldsymbol{\omega}_{R_c/N}$, we'll use an equation (3.27) in the textbook [1] :

$$[\dot{C}] = -[\tilde{\boldsymbol{\omega}}][C] \quad (3.1)$$

Because the reference-frame motion varies smoothly over time, finite-difference differentiation provides a sufficiently accurate approximation of the frame angular velocity. Therefore,

$$[\tilde{\boldsymbol{\omega}}_{R_c/N}] = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} = -[R_c \dot{N}][R_c N]^T \quad (3.2)$$

To derive the rate of a DCM numerically, we'll have:

$$[R_c \dot{N}] \approx \frac{[R_c N]_{t+dt} - [R_c N]_t}{dt} \quad (3.3)$$

This algorithm is implemented in MATLAB function `RcN_omega(t)`. The choice of different `dt` will result in different precisions in finding the angular velocity.

3.3 Attitude Evaluation and Simulator

3.3.1 Task 6: Attitude Error Evaluation

Any attitude feedback control algorithm requires the ability to compute the attitude and angular velocity tracking errors of the current body frame $\mathcal{B}$ relative to the reference frame $\mathcal{R}$. To find a set of tracking errors $\begin{bmatrix} \sigma_{B/R} \\ {}^{\mathcal{B}}\omega_{B/R} \end{bmatrix}$, we have the known variables $\sigma_{B/N}$, ${}^{\mathcal{B}}\omega_{B/N}$, as well as the desired reference frame DCM $[RN]$ and frame rate ${}^{\mathcal{N}}\omega_{R/N}$. We first construct the DCM $[BN]$ which can be obtained from $\sigma_{B/N}$ using equation (3.152):

$$[C] = [I_{3 \times 3}] + \frac{8[\tilde{\sigma}]^2 - 4(1 - \sigma^2)[\tilde{\sigma}]}{(1 + \sigma^2)^2} \quad (3.4)$$

The inverse transformation—converting a DCM to its MRP form—is formulated in equation (3.153):

$$\begin{aligned} [\tilde{\sigma}] &= \frac{[C]^T - [C]}{\zeta(\zeta + 2)} \\ \zeta &= \sqrt{\text{trace}([C]) + 1} \end{aligned} \quad (3.5)$$

These two forms are implemented in MATLAB function `sigmaToDCM()` and `DCMToSigma()` respectively.

Now, the roadmap to finding the tracking errors would be:

$$\begin{aligned} \sigma_{B/N} &\rightarrow [BN] \\ [BR] &= [BN][NR] = [BN][RN]^T \\ [BR] &\rightarrow \sigma_{B/R} \\ {}^{\mathcal{B}}\omega_{B/R} &= {}^{\mathcal{B}}\omega_{B/N} - {}^{\mathcal{B}}\omega_{R/N} = {}^{\mathcal{B}}\omega_{B/N} - [BN]{}^{\mathcal{N}}\omega_{R/N} \end{aligned}$$

A MATLAB function named `attitudeErrorEval()` was implemented to evaluate both attitude and angular velocity tracking errors.

3.3.2 Task 7: Numerical Attitude Simulator

In order to integrate the attitude we need a numerical integrator. The LMO and GMO orbit satellite positions are already calculated in earlier tasks. Let the propagated attitude

state $\mathbf{X}$ be

$$\mathbf{X} = \begin{bmatrix} \boldsymbol{\sigma}_{B/N} \\ {}^{\mathcal{B}}\boldsymbol{\omega}_{B/N} \end{bmatrix}$$

Assuming that the spacecraft is rigid and there's no external torque acting on it, its dynamics obey

$$[I]\dot{\boldsymbol{\omega}}_{B/N} = -[\tilde{\boldsymbol{\omega}}_{B/N}][I]\boldsymbol{\omega}_{B/N} + \mathbf{u} \quad (3.6)$$

where $\mathbf{u}$ is the external control torque vector.

For this task, a loop integrator was written in MATLAB. The control torques ${}^{\mathcal{B}}\mathbf{u}_1 = 0$ and ${}^{\mathcal{B}}\mathbf{u}_2 = (0.01, -0.01, 0.02)$ were applied respectively and the following were computed:

- Attitude MRP $\boldsymbol{\sigma}_{B/N}$
- Angular velocity $\boldsymbol{\omega}_{B/N}$
- Angular momentum vector $\mathbf{H} = [I]\boldsymbol{\omega}_{B/N}$ in both $\mathcal{N}$ and $\mathcal{B}$ frames
- Rotational kinetic energy $T = \frac{1}{2}\boldsymbol{\omega}_{B/N}^T [I]\boldsymbol{\omega}_{B/N}$

The results of both simulations are plotted in Figure 3.1. As is seen, after applying the control torque $\mathbf{u}_2$ on the spacecraft, its rotational kinetic energy increases significantly (from approximately 9 mJ in uncontrolled mode (Figure 3.1a) to more than 2.5 J when controlled (Figure 3.1b)). This is verified by spacecraft angular velocity, whose 1st and 2nd components start to oscillate in controlled mode while its 3rd goes through a rapid, non-stop increase (Figure 3.1d).

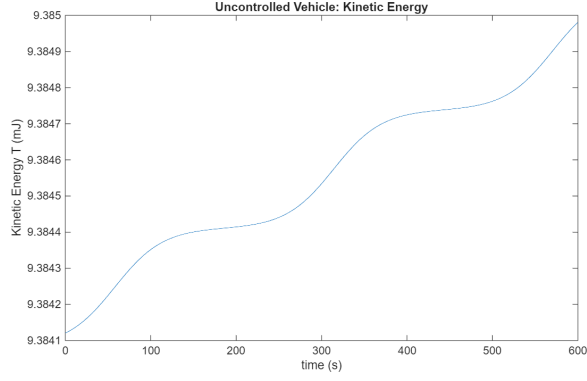
The attitude-state histories further confirm this behavior. The spacecraft exhibits unstable rotational behavior when the control torque $\mathbf{u}_2$ is applied to it (Figure 3.1f). These results indicate that the applied torque does not stabilize the spacecraft attitude dynamics.

3.4 Completing the Mission

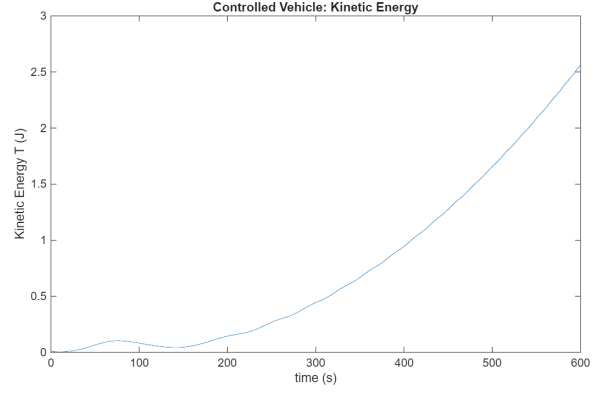
3.4.1 Task 8: Sun Pointing Control

The PD gains were selected using the linearized closed-loop attitude dynamics together with a worst-case inertia design approach. Since scalar gains were used for all spacecraft axes, the gains were chosen based on the maximum and minimum principal moments of inertia to guarantee that all rotational modes satisfy the required 120-second decay time constant.

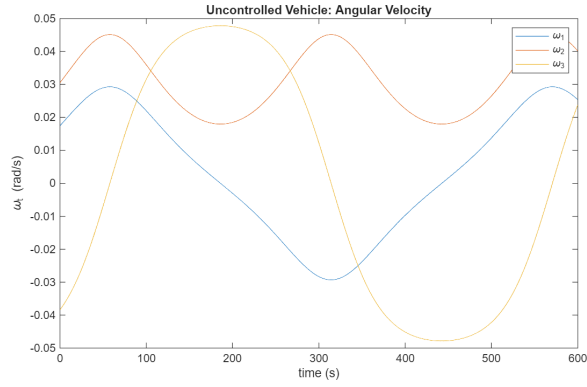
$$I_{max} = 10, I_{min} = 5$$



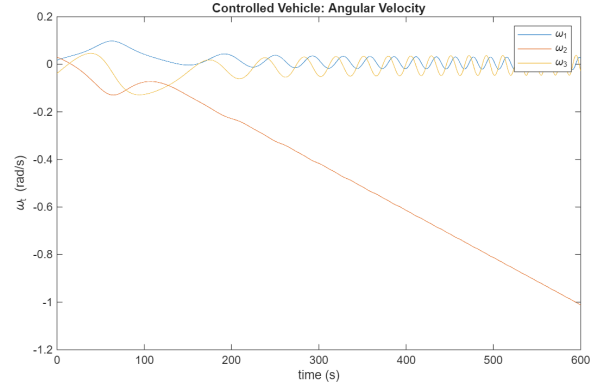
(a) u_1 applied: T



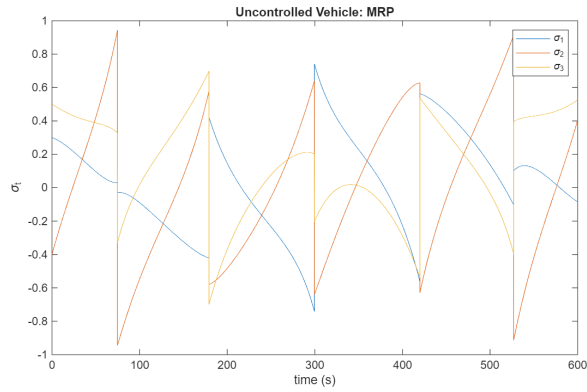
(b) u_2 applied: T



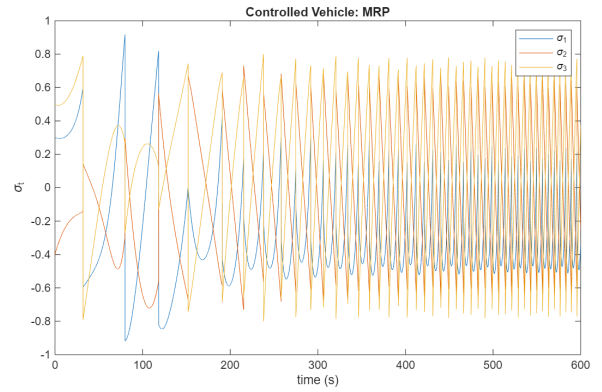
(c) u_1 applied: $\omega_{B/N}$



(d) u_2 applied: $\omega_{B/N}$



(e) u_1 applied: $\sigma_{B/N}$



(f) u_2 applied: $\sigma_{B/N}$

Figure 3.1: Results of the simulations in task 7

$$\tau = 120s$$

$$K = \frac{4I_{max}^2}{\tau^2 I_{min}}, P = \frac{2I_{max}}{\tau} \quad (3.7)$$

The control torque is defined as:

$${}^B\mathbf{u} = -K\boldsymbol{\sigma}_{B/R} - P{}^B\boldsymbol{\omega}_{B/R}$$

An RK4 integrator was developed to perform numerical propagation of the spacecraft attitude dynamics used by the MATLAB function `simulate_mode()`. The results of Sun pointing simulation are evident in Figure 3.2.

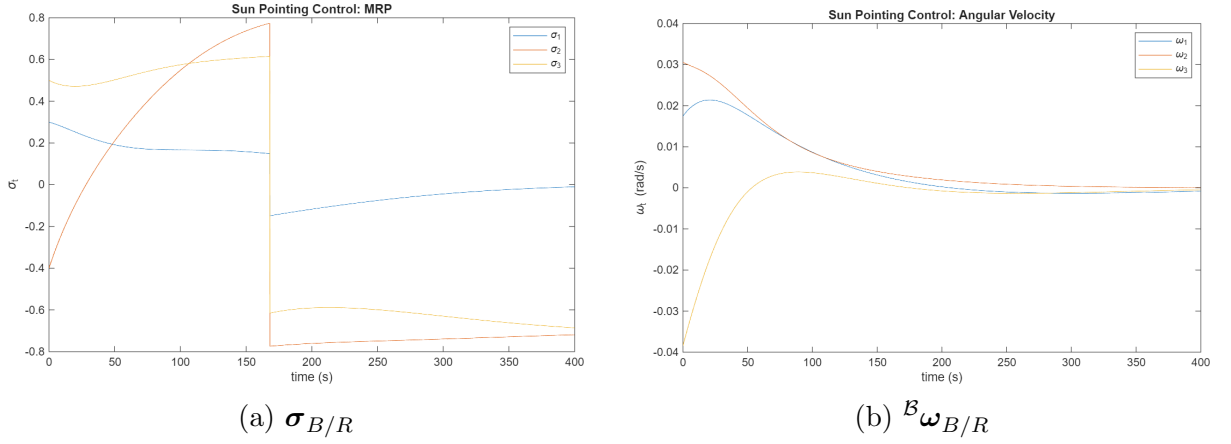


Figure 3.2: Results of Sun pointing control

As is illustrated, the spacecraft is fully detumbled (${}^B\boldsymbol{\omega}_{B/R} \approx 0$) and driven toward the desired reference attitude.

3.4.2 Task 9: Nadir Pointing Control

The same PD control torque developed in task 8 is also applied here and the results of simulation for Nadir pointing mode are evident in Figure 3.3.

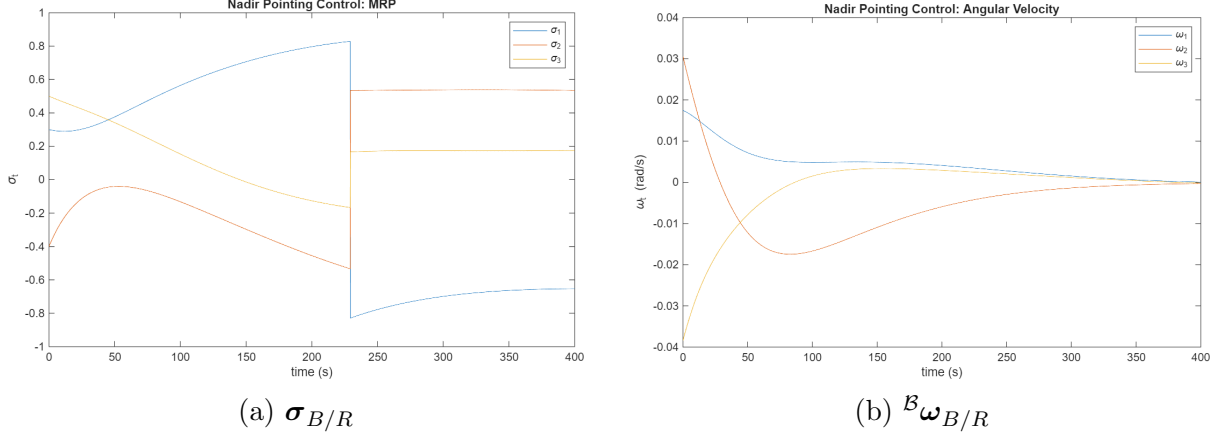


Figure 3.3: Results of Nadir pointing control

Similar to the previous task, the attitude tracking errors converge asymptotically toward zero and the spacecraft is also detumbled.

3.4.3 Task 10: GMO Pointing Control

The same PD control torque used in two previous tasks is also applied here and the results of simulation for GMO pointing mode are evident in Figure 3.4.

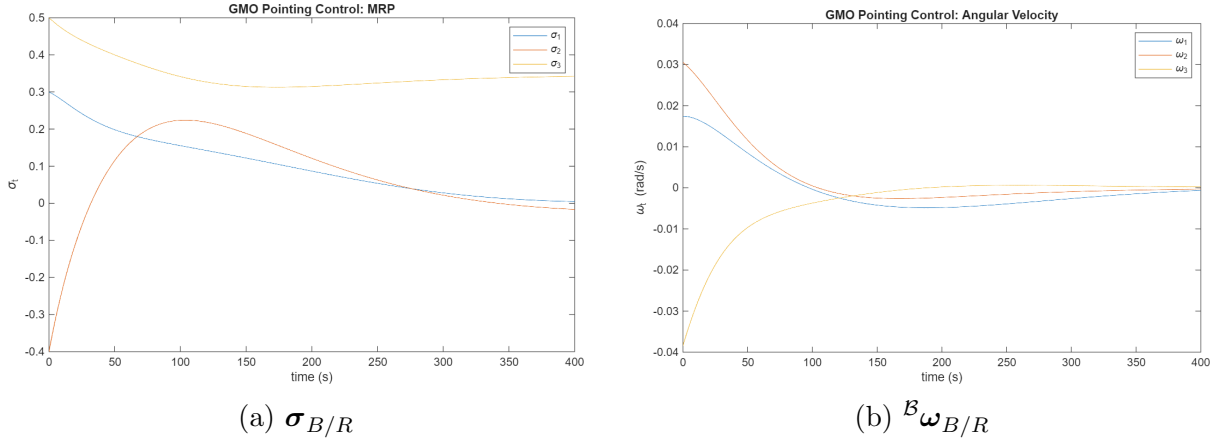


Figure 3.4: Results of GMO pointing control

3.4.4 Task 11: Mission Scenario Simulation

A complete mission simulation was developed to automatically switch between spacecraft operational modes according to mission conditions.

The mission logic was implemented as follows:

- If the spacecraft was illuminated by the Sun, Sun-pointing mode was activated,
- If the spacecraft was in eclipse and the GMO spacecraft was visible, communication mode was activated,

- Otherwise, nadir-pointing science mode was selected.

The task of finding out the operation mode at each given time in the mission—and choosing the corresponding reference frame as a result—is carried out by the MATLAB function `getReferenceFrame()`.

The MATLAB function `final_simulation()` handles the procedure of combining all the modes and tasks together in a timespan of 6500 seconds.

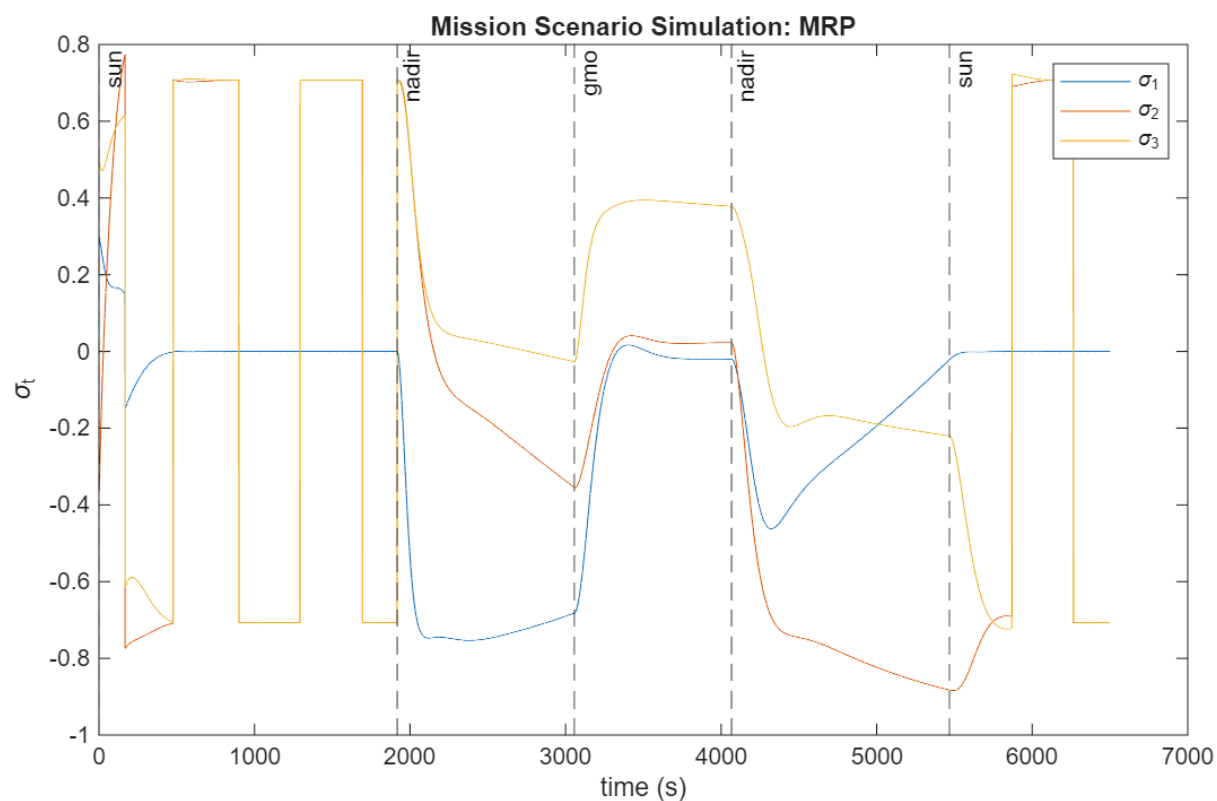


Figure 3.5: Full mission spacecraft attitude state

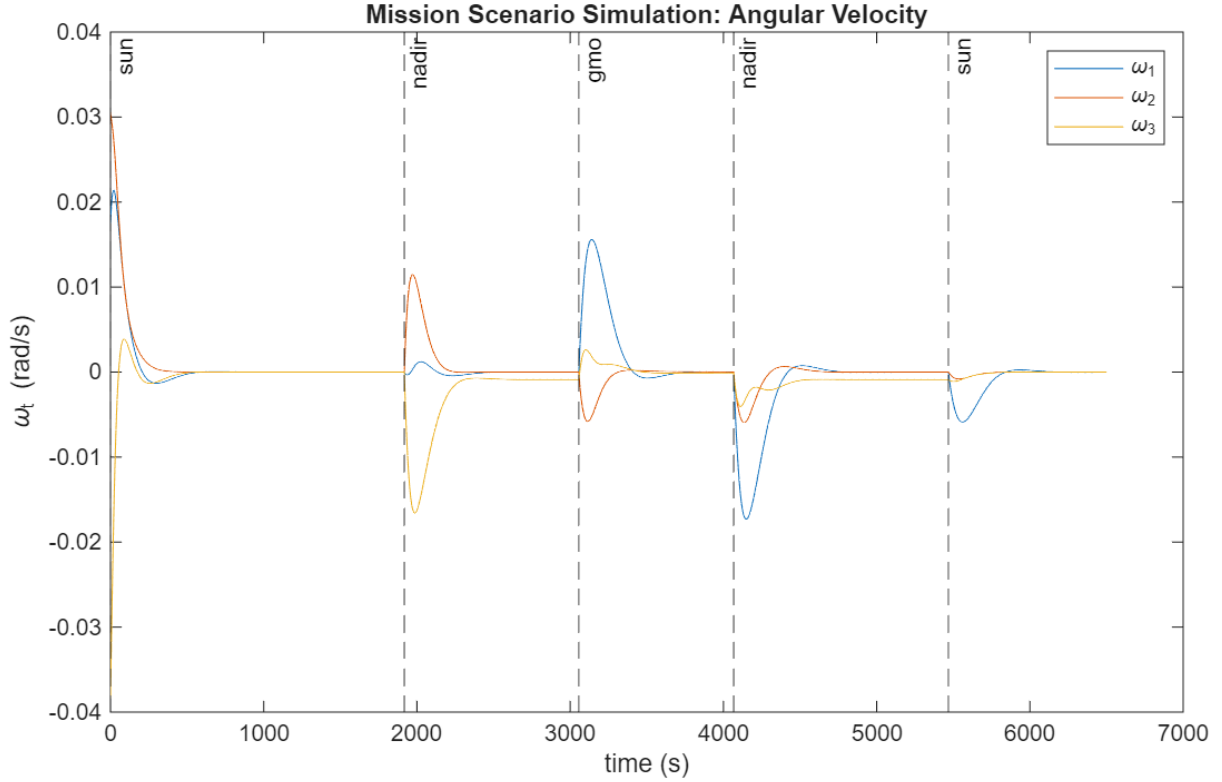


Figure 3.6: Full mission spacecraft angular velocity

3.5 Simulation Results and Discussion

The complete mission simulation demonstrates the ability of the spacecraft attitude-control architecture to autonomously transition between multiple operational pointing modes while maintaining stable closed-loop rotational behavior.

Unlike the previous tasks, where each reference frame remained fixed to a single operational objective, the reference frame in Task 11 varies dynamically according to orbital geometry and mission conditions. Consequently, the controller must continuously track time-varying reference attitudes generated by the mission-management logic.

The simulation results indicate that the spacecraft successfully transitions between Sun-pointing, communication, and nadir-pointing operational modes without loss of stability. During mode-switching events, temporary increases in attitude and angular velocity tracking errors are observed due to the sudden change in commanded reference orientation. However, the proportional-derivative feedback controller rapidly damps these transients and drives the spacecraft toward the new reference frame.

The angular velocity histories demonstrate that the spacecraft remains fully detumbled throughout the mission despite repeated reference-frame transitions. Short-duration angular-rate excursions occur immediately after mode changes as the spacecraft reorients itself toward the newly commanded operational attitude. The derivative feedback term provides sufficient damping to prevent sustained oscillatory behavior.

The operational-mode transitions are currently implemented as instantaneous reference-frame changes. In practical spacecraft systems, slew maneuvers are typically constrained by actuator saturation limits, momentum management requirements, and attitude-rate limitations. Future implementations may therefore incorporate constrained slew trajectory generation to smooth operational transitions.

Overall, the integrated simulation confirms the effectiveness of the spacecraft attitude dynamics and control architecture in performing stable autonomous mission operations under varying orbital conditions.

3.6 STK Visualization

To validate the MATLAB simulations visually, spacecraft orbit propagation and attitude motion were reconstructed using AGI Systems Tool Kit (STK).

The generated visualization confirmed:

1. Correct orbital motion of both spacecraft,
2. Proper nadir and Sun-pointing orientations,
3. Communication alignment during GMO visibility windows,
4. Stable attitude transitions throughout mission operation.

3.7 Appendix: MATLAB Function Overview

Table 3.1: Main MATLAB Functions Used in the Project

Function Name	Description
<code>orbit_state()</code>	Computes spacecraft inertial position and velocity vectors
<code>orbitTheta()</code>	Computes orbital angle evolution
<code>orbit_frame_dcm()</code>	Generates Hill-frame DCM
<code>RsN_DCM()</code>	Returns Sun-pointing reference-frame DCM
<code>RnN_DCM()</code>	Returns nadir-pointing reference-frame DCM
<code>RcN_DCM()</code>	Returns GMO-pointing reference-frame DCM
<code>attitudeErrorEval()</code>	Computes tracking errors
<code>simulate_mode()</code>	Simulates spacecraft attitude dynamics
<code>getReferenceFrame()</code>	Performs automatic mission-mode selection

Conclusion

This project presented the complete development and implementation of an attitude dynamics and control framework for a nano-satellite orbiting Mars. Multiple mission pointing modes were developed, including Sun-pointing, nadir-pointing, and GMO communication-pointing configurations.

The spacecraft dynamics, reference-frame generation, attitude tracking evaluation, and closed-loop control architecture were successfully implemented in MATLAB and validated through numerical simulation.

The final integrated mission simulation demonstrated stable spacecraft attitude behavior while autonomously switching between operational mission modes according to orbital conditions.

Future improvements may include higher-fidelity environmental disturbance modeling, actuator limitations, sensor-noise modeling, and advanced nonlinear control strategies.

Bibliography

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